[METRIC]
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PERFORMANCE SPECIFICATION

VALVE, FOUR-WAY, HYDRAULIC, MANUAL (METRIC)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of manually operated four-way hydraulic directional control valves of 20,684 kilopascals (kPa) [3000 pounds per square inch (psi)] operating pressure (see 6.1).

1.2 Classification. Valves shall be of the following types, and classifications as specified (see 6.2):

Type I - Four-Way, Two-Position, 7.6 liters per minute (L/m) [2 gallons per minute (gpm)] flow capacity.
Type II - Four-Way, Three-Position.
Class A - 7.6 L/m (2 gpm) rated flow capacity.
Class B - 22.7 L/m (6 gpm) rated flow capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2590

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for addition information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-H-5606	- Hydraulic Fluid, Petroleum Base: Aircraft, Missile and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base: For Preservation and Operation.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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FEDERAL REGULATIONS

Code of Federal Regulations

(Copies of the Code of Federal Regulations may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

DEPARTMENT OF DEFENSE

9093264	- Valve, Two-Position, Four-Way.
9094352	- Valve, Three-Position, Four-Way.
9098279	- Valve, Three-Position, Four-Way.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4 - Sampling Procedures and Tables for Inspections by Attributes.

(Copies of ASQC Standards may be obtained from The American Society for Quality Control, P.O. Box 305, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

GENERAL MOTORS CORPORATION

GM 9540P - Accelerated Corrosion Test

(Copies of GM 9540P may be obtained from the Global Engineering, 7730 Carondelet Avenue, Suite 407, St. Louis, MO 63105.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for hydraulic valves. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.5.1). Recovered materials shall be used to the maximum extent practicable. Asbestos, cadmium, and radioactive material will not be used in any form or part of this item. Radioactive material is defined by 1) Title 10, Code of Federal Regulations, Part 40, and 2) Other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.3 Design and construction. Unless otherwise specified (see 6.2), the valve shall be constructed to the form and dimensions shown on the applicable drawing listed in table I (see 4.5.2). The shape is optional within the maximum dimensions.

TABLE I. Applicable drawings.

Valve Type	Valve Class	Drawing Number
I	-	9093264
II	A	9098279
	B	9094352

3.3.1 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Subassemblies, composed of mating components, must be interchangeable as assembled units (see 4.5.2).

3.3.2 Valve spool. Valve spool stems exposed to dirt and moisture shall be provided with wipers and seals meeting the life requirements specified herein. All exposed control valve spool surfaces shall be hardened to Rockwell C 50 to 65. The spool surface shall be plated to resist corrosion. After plating, maximum spool surface roughness shall not exceed 16 micro-inches.

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The spool shall be hydraulically balanced so that forces required to manipulate the levers shall not vary more than 10 percent (%) due to variations in pressure and flow. Spools shall not bind under any operating conditions specified herein (see 4.5.2).

3.4 Performance.

3.4.1 Rated flow and pressure drop.

3.4.1.1 Type I. The rated flow shall be 7.6 L/m (2 gpm) through any interconnected ports with no more than 1034 kPa (150 psi) pressure drop across the valve (see 4.5.4.1).

3.4.1.2 Type II. The rated flow of class A valves shall be 7.6 L/m (2 gpm) through any interconnected ports, with less than 689 kPa (100 psi) pressure drop across the valve. The rated flow of class B valves shall be 22.7 L/m (6 gpm) with less than 1724 kPa (250 psi) pressure drop across the valve (see 4.5.4.1).

3.4.2 Operating torque.

3.4.2.1 Type I. The maximum operating torque shall be 5.6 Newton meters (N.m) [50 pound-inches (lb in.)] with 20 684 kPa (3000 psi) applied at the pressure port (see 4.5.4.2).

3.4.2.2 Type II. The maximum operating torque shall be 2.8 N.m (25 lb in.) For class A valves, and 5.9 N.m (52 lb in.) for class B valves, with 20 684 kPa (3000 psi) hydraulic pressure applied at the pressure port (see 4.5.4.2).

3.4.3 Operating pressure. The valve shall operate satisfactorily throughout a hydraulic pressure range of 0 to 26 684 kPa (3000 psi) applied to any port, except the return port, while using hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083 or the commercial equivalent (see 4.5.4.3).

3.4.4 Proof pressure. The valve shall withstand, for 2 minutes, a hydraulic proof pressure of 31 026 kPa (4500 psi), except for the return port which shall withstand 3447 kPa (500 psi) for type I valves and 10 342 kPa (1500 psi) for type II valves without external leakage, permanent deformation, or failure (see 4.5.4.4).

3.4.5 Burst pressure. The valve shall withstand, for 3 minutes in each position, a hydraulic pressure of 51 711 kPa (7500 psi), except for the return port which shall withstand 5723 kPa (830 psi) for type I valves and 17 237 kPa (2500 psi) for type II valves (see 4.5.4.5).

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3.4.6 Leakage. The valve shall withstand application of 20 684 kPa (3000 psi) hydraulic pressure to sealing members of the individual fluid chambers without external leakage, and with a combined internal leakage from the ports not exceeding 10 cubic centimeters (cc) per minute (see 4.5.4.6).

3.4.7 Life. The valve shall withstand 20 000 cycles of operation at 20 684 kPa (3000 psi) hydraulic working pressure, without external leakage, and with subsequent performance conforming to requirements of 3.4.6 (see 4.5.4.7).

3.5 Environmental requirements.

3.5.1 High temperature. The valve shall perform in accordance with requirements of 3.4.1 and 3.4.7 while operating at 71 ± 3 degrees Celsius (°C) [160 ± 5 degrees Fahrenheit (°F)]. Subsequently, the valve shall conform to requirements of 3.4.1, 3.4.6, and 3.4.7 (see 4.5.5.1).

3.5.2 Low temperature. The valve shall perform in accordance with requirements of 3.4.1 and 3.4.7 while operating at -54 ± 3°C (-65 ± 5°F). Subsequently, the valve shall conform to requirements of 3.4.1, 3.4.6 and 3.4.7 (see 4.5.5.2).

3.5.3 Fungus resistance. The valve shall evidence no microbial growth that would adversely affect performance after 28 days exposure to a spore incubation as specified in 4.5.5.3.

3.5.4 Corrosion resistance. The valve and the removed valve spool shall evidence no corrosion after testing as specified in 4.5.5.4.

3.5.5 Dust resistance. The valve shall operate after 6 hours exposure to blowing dust as specified in 4.5.5.5.

3.5.6 Shock resistance. The valve shall operate after exposure to a shock impulse as specified in 4.5.5.6.

3.5.7 Vibration resistance. The valve components shall not loosen nor become damaged following the vibration test specified in 4.5.5.7.

3.6 Weight. The gross weight of the valve shall not exceed 0.9 kilograms (kg) (2.0 pounds) for type I valves and 1.1 kg (2.5 pounds) for type II valves (see 4.5.2).

3.7 Finish. The finish shall be as specified in the contractor order (see 4.5.2 and 6.2).

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3.8 Identification and marking. Markings shall be permanent and legible. When specified in the contract or order (see 6.2), nameplates shall be included. Nameplates shall be made of a material suitable to withstand the environmental requirements specified herein and remain legible. Marking shall include the following information in the order shown (see 4.5.2):

Valve, 4-way, hydraulic, manual
National Stock Number (NSN)
Part or identifying number (PIN)
Manufacturer's Serial Number
Manufacturer's name, identification code (CAGE) or trademark
Contract or order number
Date of manufacture
US

3.9 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, or other defects which may adversely affect the serviceability or appearance of the valve (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1).

4.2.1 Preproduction inspection. When specified (see 6.2), first article preproduction inspection shall be performed on four valves. This inspection shall include the examinations of 4.4 and the tests of 4.5.4.1 through 4.5.5.7 (see table II).

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), first article preproduction inspection shall be performed on four valves, from the first 10 valves produced under the Government contract. This examination shall include the examinations of 4.4 and the tests of 4.5.4.1 through 4.5.5.7 (see table II).

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TABLE II. Classification of inspections.

Title	Requirement	Inspec-tion	First article	Quality conformance	
				Exami-nation	Tests
Materials and Defects (see table II	3.2 3.3 thru 3.3.2, 3.6 thru 3.9	4.5.1 4.5.2	X X	X	
Rated flow and pressure drop					
Type I	3.4.1.1	4.5.4.1	X		X
Type II	3.4.1.2	4.5.4.1	X		X
Operating torque					
Type I	3.4.2.1	4.5.4.2	X		X
Type II	3.4.2.2	4.5.4.2	X		X
Operating pressure	3.4.3	4.5.4.3	X		X
Proof pressure					
Type I	3.4.4	4.5.4.4	X		
Type II	3.4.4	4.5.4.4	X		
Burst pressure	3.4.5	4.5.4.5	X		
Leakage	3.4.6	4.5.4.6	X		X
Life	3.4.7	4.5.4.7	X		
High temperature	3.5.1	4.5.5.1	X		
Low temperature	3.5.2	4.5.5.2	X		
Fungus resistance	3.5.3	4.5.5.3	X		
Corrosion resistance	3.5.4	4.5.5.4 4.5.5.4.1	X X		
Sand and dust resistance	3.5.5	4.5.5.5	X		
Shock resistance	3.5.6	4.5.5.6	X		
Vibration resistance	3.5.7	4.5.5.7	X		

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.5.4.1, 4.5.4.2, 4.5.4.3 and 4.5.4.6 (see table II).

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4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ASQC Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements of 3.3, 3.3.1, 3.3.2, and 3.6 through 3.9 as listed in table III. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3, 3.3.1, 3.3.2 and 3.6 through 3.9 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with SIE.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Dimensions affecting function and interchangeability not in accordance with applicable drawing (see 3.3 and 3.3.1).	Gage or functional
102	Valve spool stem wipers or seals missing or defective (see 3.3.2).	Visual
103	Valve spool hardness not as specified (see 3.3.2).	SIE 1/
104	Valve spool surface finish in excess of that specified (see 3.3.2).	SIE
105	Hydraulic balance of spool not within limits (see 3.3.2).	Functional
106	Binding (see 3.3.2).	Functional

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TABLE III. Classification of defects - Continued.

Category	Defect	Method of examination
Minor		
201	Weight greater than specified (see 3.6) .	Scale
202	Finish not as specified (see 3.7).	Visual
203	Marking illegible or incorrect (see 3.8).	Visual
204	Faulty workmanship (see 3.9).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Test conditions. Unless otherwise specified (see 6.2), ambient air temperature shall be 23 ± 10°C (77 ± 15°F), barometric pressure shall be 725 +50 -75 mm mercury (Hg) (28.5 +2 inches Hg), and relative humidity shall be 50 ± 30 percent. Unless otherwise specified herein, the valve shall meet all specified performance requirements in a hydraulic system, utilizing hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083 or the commercial equivalents under the following conditions:

a. Hydraulic fluid temperature of 38 ± 5°C (70 - 100°F).
b. All grease or foreign matter shall be removed from the interior parts of the valve before testing.
c. Air shall be completely removed from all hydraulic portions of the valve prior to test.

4.5.4 Performance.

4.5.4.1 Rated flow and pressure drop test. The rated flow of the valve shall be accurately maintained at the applicable value specified in 3.4.1.1 for type I and 3.4.1.2 for type II valves. A manometer or other accurate means of measuring pressure drop from one port to another, shall be employed. The pressure drop for the valves shall be recorded to determine conformance to 3.4.1.1 and 3.4.1.2.

4.5.4.2 Operating torque test. With a hydraulic pressure of 20 684 kPa (3000 psi) at the pressure port, the torque required to operate the valves shall be measured and recorded to determine conformance to 3.4.2.1 and 3.4.2.2.

4.5.4.3 Operating pressure test. Pressure shall be applied to each port, except the return port, and varied throughout a range of 0 psi to 20 684 kPa (3000 psi). Operation shall be observed to determine conformance to 3.4.3.

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4.5.4.4 Proof pressure. Proof pressure in accordance with 3.4.4 shall be applied at the pressure and return ports for a period of 2 minutes. The valve shall be manually actuated after application of pressures. There shall be no evidence of external leakage, permanent deformation or failure. The results shall be recorded to determine conformance to 3.4.4.

4.5.4.5 Burst pressure. This test shall be performed after completion of all other tests. The valve shall be subjected to the values of pressure specified in 3.4.5 for the pressure ports and return port for a period of 3 minutes at each application. This test shall be performed for all three positions. When testing pressure ports, the return port shall be open. The rate of application of pressure shall not exceed 179 267 kPa (26 000 psi) per minute. The valve need not be operable after these tests. Results shall be recorded to determine conformance to 3.4.5.

4.5.4.6 Leakage. The valve shall be tested for leakage by applying a hydraulic pressure of 20 684 kPa (3000 psi) to each sealing member of individual fluid chambers, in succession for a 2-minute period each. The pressure shall then be lowered to 34 to 69 kPa (5 to 10 psi) (gauge) and held for 2 minutes. There shall be no external leakage and the combined internal leakage from each port shall be recorded to determine conformance to 3.4.6.

4.5.4.7 Life test. The valve shall be subjected to a hydraulic pressure of 20 684 kPa (3000 psi) and a rated flow as specified (see 3.4.1), for 20 000 cycles. Each cycle shall consist of operating the valve from one extreme position to the other extreme position and return to the starting position. The valve shall be observed for evidence of external leakage during cycling. After cycling, internal leakage shall not exceed the value specified in 3.4.6. The value shall evidence no excessive wear or breakdown and the results shall be recorded to determine conformance to 3.4.7.

4.5.5 Environmental tests.

4.5.5.1 High temperature. The valve shall be placed within a test chamber and exposed to a temperature of 71°C (160°F) for 24 hours duration. After 24 hours of exposure, the valve shall be charged to 20 684 kPa (3000 psi) hydraulic pressure and rated flow (see 3.4.1) which shall be maintained for a period of 4 hours. At the conclusion of the exposure period, the valve shall be tested as specified in 4.5.4.7 for 50 cycles. The valve shall then be cooled to room temperature and tested as specified in 4.5.4.1, 4.5.4.6 and 4.5.4.7 for 50 cycles to determine conformance to 3.5.1.

4.5.5.2 Low temperature test. The valve shall be placed within a test chamber and exposed to a temperature of -54°C (-65°F) for 24 hours duration. After 24 hours of low temperature exposure, the valve shall be charged to a hydraulic pressure of 20 684 kPa (3000 psi), and a rated flow as specified (see 3.4.1) shall be maintained for a period of 4 hours. At the

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conclusion of the 4-hour period, the valve shall be tested as specified in 4.5.4.7 for 50 cycles. The valve shall then be stabilized at room temperature and tested as specified in 4.5.4.1, 4.5.4.6, and 4.5.4.7 for 50 cycles to determine conformance to 3.5.2.

4.5.5.3 Fungus resistance test. The valve shall be subjected to the fungus resistance test specified in method 508 of MIL-STD-810. Subsequent to this test the valve shall be tested as specified in 4.5.4.1, 4.5.4.6 and 4.5.4.7 for 50 cycles to determine conformance to 3.5.3.

4.5.5.4 Corrosion resistance test. The valve assembly shall be subjected to 40 cycles of Accelerated Corrosion Test GM 9540P, method B. Following the exposure the valve shall be tested as specified in 4.5.4.1, 4.5.4.6, and 4.5.4.7 for 50 cycles to determine conformance to 3.5.4.

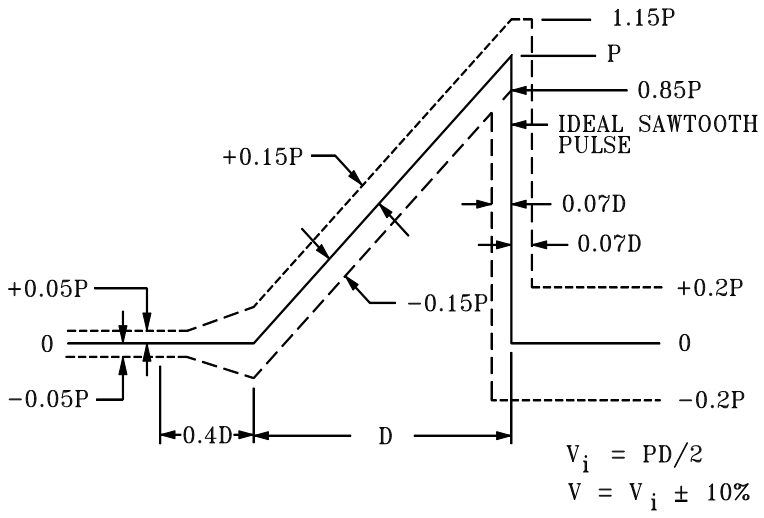
4.5.5.4.1 Valve spool corrosion test. The valve spool shall be removed from the assembly, tested as specified in 4.5.4.1, 4.5.4.6, and 4.5.4.7 and subjected to 20 cycles of Accelerated Corrosion Test GM 9450P, method B. After testing, the valve spool shall exhibit no corrosion in order to meet the corrosion requirement of 3.5.4.

4.5.5.5 Dust resistance test. The valve shall be subjected to the blowing dust test (procedure I) specified in method 510 of MIL-STD-810, except that the maximum temperature shall be 52°C (+125°F). Following the test, the valve shall be tested as specified in 4.5.4.1, 4.5.4.6, and 4.5.4.7 for 50 cycles to determine conformance to 3.5.5.

4.5.5.6 Shock resistance test. The valve shall be subjected to a half sine shock pulse of 15 gravity units (g) for a period of 11 + milliseconds (see figure 1). Following the test, the valve shall be examined for evidence of damage and tested as specified in 4.5.4.1, 4.5.4.6, and 4.5.4.7 for 50 cycles to determine conformance to 3.4.6.

4.5.5.7 Vibration test. The valve shall be subjected to a simple harmonic motion having an amplitude of 0.8 millimeter (mm) (0.03 inch) [1.5 mm (0.06 inch) maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 hertz (Hz). The entire frequency range, from 10 to 55 Hz and return to 10 Hz, shall be traversed in approximately 1 minute. The vibration shall be for 1 hour at 25 $\pm$ 8°C (77 $\pm$ 15°F) and 30 minutes at -54°C (-65 $\pm$ 5°F) in each direction of the three major axes. Following the vibration, the valve shall be examined for evidence of damage and tested as specified in 4.5.4.1, 4.5.4.6, and 4.5.4.7 for 50 cycles to determine conformance to 3.5.7.

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NOTE:
The oscillogram should include a time about 3D long with the pulse located approximately in the center. The integration to determine velocity change should extend from .4D before the pulse to .1D beyond the pulse. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. Any measured acceleration pulse which can be contained between the broken line boundaries is a nominal half sine pulse of nominal amplitude A and nominal duration D. The velocity-change associated with the measured acceleration pulse is V.

FIGURE 1. Tolerances for half sine shock pulse.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

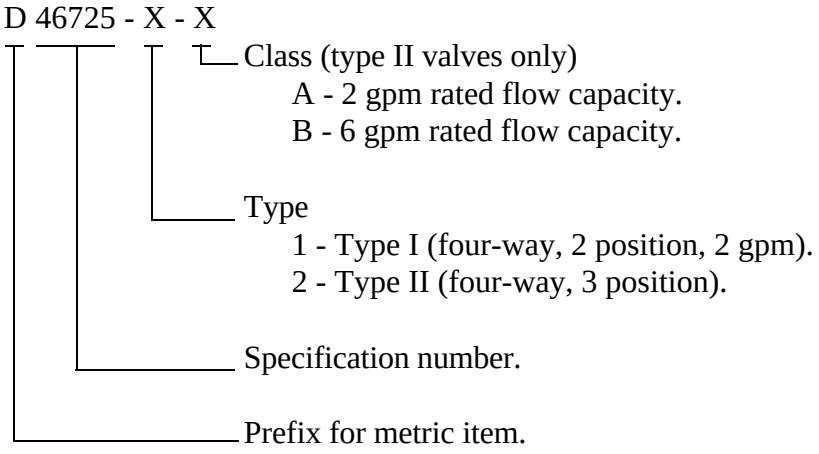
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6.1 Intended use. The valves covered by this performance specification are intended to be used to manually select the fluid flow paths in a 3000 psi hydraulic system in order to control cylinder actuation. They are intended for use with the loader-transporter, guided missile (Hawk-Loader).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Type and class of valve required (see 1.2).
d. If first article samples are not required (see 3.1).
e. If valve construction is other than as specified (see 3.3).
f. Valve finish (see 3.7).
g. Nameplate, if required (see 3.8).
h. If preproduction inspection is required (see 4.2.1).
i. If initial production inspection is not required (see 4.2.2).
j. Test conditions, if other than as specified (see 4.5.3).
k. Packaging requirements (see 5.1).
l. PIN and quantity required (see 6.3).

6.3 Part or identifying number (PIN). The pins to be used for valves acquired to this specification are established as follows:



6.4 Subject term (key word) listing.

Four-way hydraulic valves
Hydraulic valve, three-position, four-way manual
Hydraulic valves, four-way, manual

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Manual hydraulic valve, three-position, four-way
Manually operated valves

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2590-0237)

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